

Social Media Engagement Analytics — One Page Summary

Project Overview:

This project analyzes a social media engagement dataset containing information about posts, users, and engagement metrics such as likes, comments, shares, impressions, watch time, followers, and engagement rate. The objective is to clean the data, explore patterns, visualize trends, and generate insights about content performance and user behavior.

1. Descriptive Analysis:

The dataset was analyzed using Python Pandas, NumPy, Matplotlib, and Seaborn.

- The dataset contains different post types such as **image, reel, text, and video**.
- Missing values were handled using appropriate methods like **mode and median imputation**.
- Duplicate records were identified and removed.
- Data types were corrected, and categories like gender and sentiment were standardized.
- A new feature **hashtag_count** was created by extracting the number of hashtags from each post.
- Engagement score was created using likes, comments, and shares.

Key Findings:

- **Reel posts have the highest number of posts**, showing that reels are frequently used.
- **Food category content received the highest average likes**, indicating strong audience interest.
- Sentiment categories (positive, negative, neutral) showed similar follower distributions.
- Engagement metrics such as likes, comments, and shares have a positive relationship with impressions.

2. Diagnostic Analysis:

The analysis was performed using correlation, groupby summaries, and visualizations.

- Posts with higher impressions generally received more likes and engagement.
- Categories with visually attractive or popular content types tend to achieve better engagement.
- Different devices showed variations in engagement rate and user interaction.
- Sentiment analysis showed that audience response is not only dependent on sentiment but also on content quality and reach.

Insights:

- High-performing posts are influenced by:
 - Content category
 - Post type
 - Audience interest
 - Reach and impressions

3. Predictive Analysis:

Based on the patterns observed:

- Content creators can predict that **high-quality reels and popular categories like food content may generate better engagement.**
- Increasing impressions through better content strategy may improve likes and interactions.
- Hashtag usage can help improve content discoverability.
- Monitoring engagement rate can help identify future high-performing posts.

Future Recommendations:

- Focus on creating more engaging reel content.
- Analyze audience preferences before posting.
- Use effective hashtags to improve reach.
- Track engagement trends regularly to optimize content strategy.

Visualization Summary:

"The visualizations including scatter plot, line chart, bar chart, pie chart, histogram, box plot, violin plot, pair plot, heatmap, and swarm plot helped identify trends and relationships among engagement metrics."

Conclusion:

The analysis provides insights into social media performance by identifying successful content types, engagement patterns, and user behavior. These findings can help businesses and creators improve their content strategy and increase audience engagement.

Dashboard:

Social Media Engagement Analytics

☐ 2022
☐ 2023

Total Followers
2bn

Total Posts
5K

50.53M

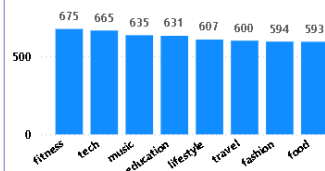
Total Comments
8M

Total Shares
5M

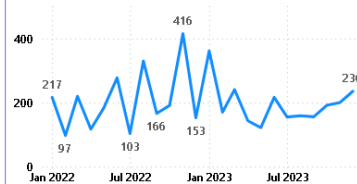
All

All

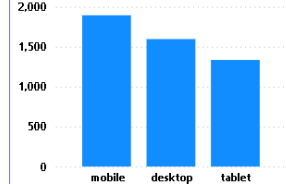
Posts by Category



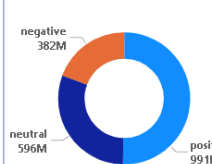
Daily Engagement Trend



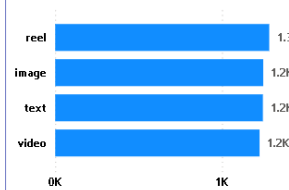
Engagement vs Device



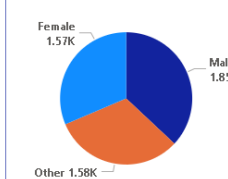
Followers vs Sentiment



Post Type Count



Gender Distribution



Average Likes by Category

